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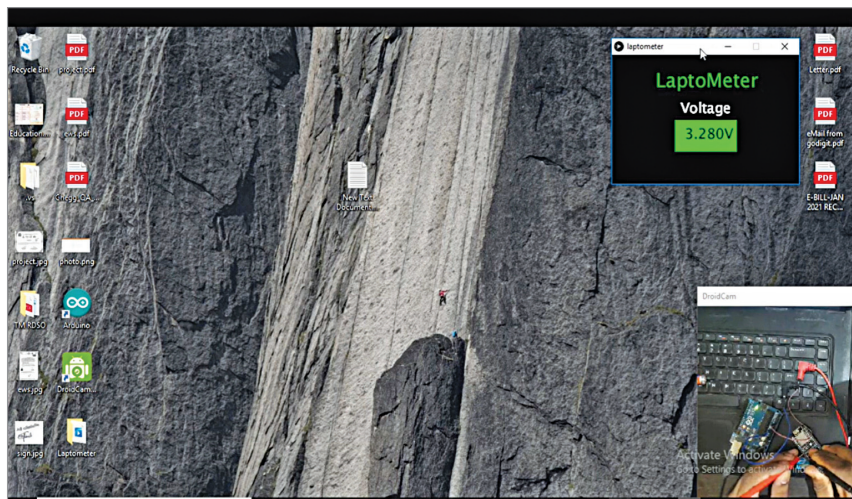


Fig. 4: Layout display on desktop

The input voltage is processed by Arduino Uno and sent to laptop through serial communication. The code in laptop reads the voltage and displays on its monitor.

The usual multimeter probes may be used for connection to A0 and GND pins of Arduino as shown in Fig. 2.

The Processing software used for coding is a flexible sketchbook and a language for learning to code within the context of the visual arts. We will write the program using this language to receive voltage values from serial port. Part of the Processing code to receive the value from serial port is given below.

```
void serialEvent(Serial my_port)
{
    value = float(my_port.
    readStringUntil('\n'));
}
```

The user interface layout to display voltage using Processing code is shown in Fig. 3. You can place the layout on the laptop's Desktop as shown in Fig. 4.

Arduino code *efyLaptoM.ino* was used for the author's prototype. It must be compiled and uploaded to Arduino Uno using Arduino IDE. In the layout designing part, Processing code *efydeslaptoM.pde* is used for this project.

Construction and testing

Connect Arduino IDE to laptop through any available COM port using a USB cable. Identify the exact COM port from the Device Manager. Then open the *efyLaptoM.ino* Arduino

code. This code must be compiled and uploaded to Arduino Uno using Arduino IDE.

Next, design the user interface layout using Processing software. Download Processing software, extract the files and click on *processing.exe* file. Processing environment will open. Open *efydeslaptoM.pde* to run it. Make sure that the COM port used in Processing should be the same as used in Arduino code. (In the prototype Arduino was connected to laptop through COM6, and the same was used in Processing code as well.)

Before using the Processing application, make sure your Arduino is programmed and connected to laptop through COM port. When Processing is run, a Dialog box will pop up as shown in Fig. 3.

Now, to measure any DC voltage up to 5V, use multimeter probes P1 and P2 at specific test points with respect to ground.

This arrangement can be used as a USB multimeter when you do not have a proper multimeter at home, though it is not handy enough to be carried around. However, it has the scope to expand its features to measure current and resistance as well like a normal multimeter. **EFY**

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